# OUEG-RAG: Controlled Evidence Graphs with Diagnostic-Guided Reasoning for Grounded Table–Text Question Answering

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**Abstract.** Answering multi-hop questions over heterogeneous semi-structured tables and unstructured text passages requires preserving tabular relational bindings while connecting cross-modal evidence from external passages. Existing retrieval-augmented generation (RAG) paradigms face an unresolved dilemma: linearizing tables into plain text destroys header hierarchies and intra-row bindings, while recent graph-based methods either prune evidence aggressively into isolated fragments—causing irreversible peripheral evidence loss—or rely on expensive, unbounded multi-turn agentic exploration over persistent knowledge bases. We propose OUEG-RAG, an online unified evidence graph framework centered on *controlled evidence scaffolding and diagnostic-guided reasoning*. At query time, OUEG-RAG constructs a bounded, schema-preserving multigraph in $< 20$ ms using star-topology record centroids to enforce horizontal row bindings alongside deterministic text triples. Rather than destructive pruning, the framework preserves complete evidence context via compact subgraph serialization while supplying deterministic multi-factor scored paths as non-filtering reading scaffolds. Crucially, OUEG-RAG establishes a controlled, bounded two-tier execution protocol: confident queries yield an immediate verified span (Fast Exit), while ambiguous candidate collisions or missing links emit a structured diagnostic handoff that triggers targeted symmetric bridge retrieval, resolving bottlenecks through factual evidence rather than heuristic tiebreaking. Extensive evaluations on the HybridQA benchmark demonstrate that OUEG-RAG consistently outperforms direct prompting, flat retrieval, and state-of-the-art graph baselines (ODYSSEY) across 20B and 120B open-weights models (+2.67% EM on 120B) while slashing empty answer rates to 0.53% and delivering 92.0% faithful provenance chains.

**Keywords:** Table question answering · Knowledge graphs · Retrieval-augmented generation · Controlled reasoning · Multi-hop grounding

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## 1 Introduction

Question answering over heterogeneous semi-structured tables and linked text passages requires navigating disparate data modalities [23]. In the HybridQA benchmark [4], answering a multi-hop query requires locating an entity within a table cell, traversing its Wikipedia hyperlink to an external unstructured passage (or vice-versa), and extracting the grounded answer span. In this setting, aligning row-level attributes, hierarchical column headers, and passage relations is essential to avert unfaithful hallucinations and relational disconnects.

The predominant baseline in retrieval-augmented generation (RAG) [13, 9, 8] is Direct In-Context Prompting, where serialized table rows and retrieved passages are supplied directly to a frozen large language model (LLM). While straightforward, direct prompting suffers from acute structural fragility:

1. *Loss of Tabular Bindings*: Standard serialization flattens tables into unstructured text, decoupling cell values from their originating row context and causing cross-row attribute conflation.
2. *Unguided Attention and Structural Ambiguity*: In complex tables, multiple candidate entities across symmetric rows frequently satisfy coarse relational constraints equally well. Under unguided prompting, LLMs struggle to disambiguate these structural ties, resorting to arbitrary guessing or hallucinated attributes.

To impart structure, recent graph-augmented methods construct knowledge graphs over tabular and textual elements [20, 14, 15]. However, existing graph architectures fall into two polar extremes: On one end, zero-shot hybrid graph systems such as ODYSSEY [1] apply *question-driven destructive pruning*, discarding up to 53% of graph nodes and edges. While reducing prompt tokens, this pruning causes severe *evidence loss*: peripheral attributes (e.g., qualifying dates, locations, or secondary conditions) present elsewhere in the row or passage are irreversibly discarded, and the model remains defenseless when multiple candidates tie. On the other end, agentic GraphRAG frameworks such as HydraRAG [18] employ unbounded multi-turn LLM agent loops and tri-factor verification over persistent global knowledge graphs (e.g., Wikidata). While expressive, these systems incur prohibitive inference latency and API costs (dozens of LLM calls per query) and cannot ingest ad-hoc, raw semi-structured tables at query time.

To resolve this dilemma, we propose **OUEG-RAG** (**O**nline **U**nified **E**vidence **G**raphs for **RAG**), organized around the core philosophy of *lossless evidence preservation coupled with controlled reasoning guidance*. Rather than using graphs as destructive filters or unbounded agent playgrounds, OUEG-RAG reframes the online multigraph as an active, auditable reasoning scaffold that disciplines and steers LLM attention.

At query time, a two-stage retriever anchors candidate table rows and expands linked Wikipedia passages. A schema-preserving multigraph is dynamically instantiated in $< 20\,\text{ms}$ (using 0 LLM calls) with star-topology record centroids $\rho_i$ that horizontally bind row attributes to column qualifiers, prevent-

ing cross-row conflation. Rather than pruning the graph, OUEG-RAG preserves 100% of the retrieved evidence via compact prompt compression while enumerating grounded multi-hop traversal paths as *non-filtering reading scaffolds* scored by a deterministic multi-factor objective $S(p)$.

Crucially, OUEG-RAG introduces an *adaptive, bounded two-tier controlled execution protocol*:

- *Tier 1 (L1 Self-Audit & Fast Exit)*: The backbone LLM reads the complete evidence multigraph guided by the top-scored path $p_1$. Under a strict controlled decoding protocol, if the evidence conclusively proves an answer, the model emits an immediate verified span (Fast Exit in a single turn). If candidate score collisions ($|\mathcal{C}_q| > 1$) or missing bridge links are detected, the model is strictly prohibited from guessing; instead, it outputs a structured *diagnostic handoff packet* specifying competing candidates, blocker types, and exact checks required.
- *Tier 2 (Diagnostic-Triggered Targeted Retrieval & L2 Resolution)*: The diagnostic handoff triggers targeted, symmetric retrieval of missing bridge passages exclusively for the focus candidates. This delta evidence is packaged into a minimal sufficient *L2 State Capsule*, allowing the L2 reader to resolve the bottleneck through concrete factual evidence rather than heuristic tiebreaking.

Evaluated on the 1,500-example stratified development benchmark of HybridQA across open-weights model scales (GPT-OSS 20B and 120B), OUEG-RAG achieves consistent gains over direct prompting, flat retrieval, and ODYSSEY (+2.67% EM on 120B, $p < 0.01$), while slashing the empty answer rate to 0.53% and delivering 92.0% faithful provenance chains.

We articulate four principal contributions:

- **Schema-Preserving Online Multigraph Construction**: A deterministic, query-time graph instantiation procedure combining star-topology record centroids $\rho_i$, qualified header predicates, and rule-based text triples in $<$ 20 ms without offline pre-indexing or LLM extraction costs.
- **Controlled Evidence Scaffolding**: A reasoning paradigm that reconciles complete evidence multigraph preservation with non-filtering multi-factor path guidance ($S(p)$), steering LLM attention without evidence loss.
- **Adaptive Two-Tier Diagnostic Grounding**: A bounded execution protocol coupling L1 self-auditing with diagnostic-triggered targeted bridge retrieval, resolving structural collisions through factual evidence rather than heuristic tiebreaking.
- **Comprehensive Empirical Verification**: Rigorous comparative evaluations against matched baselines and systematic knockout ablations, corroborated by a 100-case human provenance audit.

## 2 Related Work

*Hybrid Table–Text Question Answering.* Multi-hop reasoning over heterogeneous tabular and textual knowledge was catalyzed by HybridQA [4], which introduced

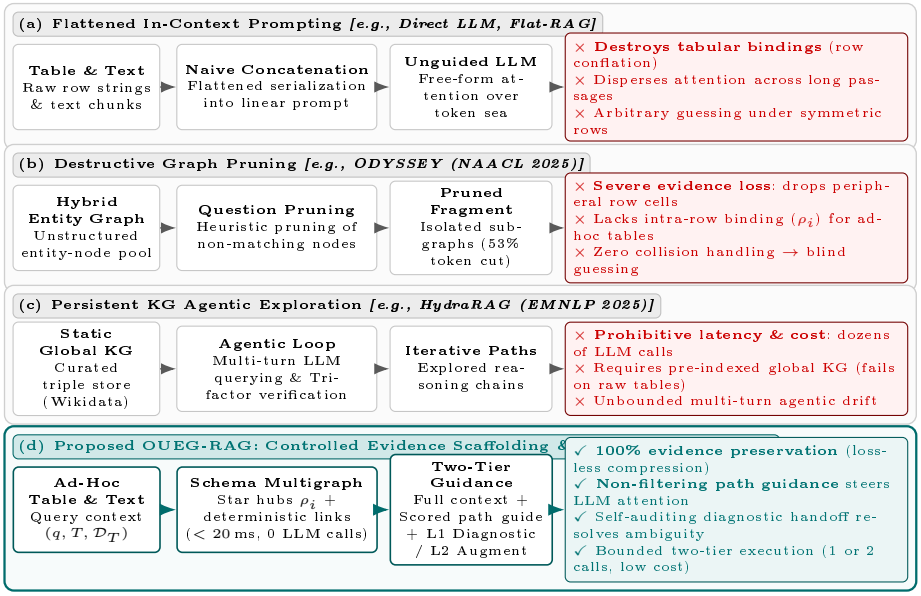



**Fig. 1.** Architectural comparison across four table-text reasoning paradigms. In contrast to unguided linear prompt concatenation (a), destructive question-driven graph pruning in ODYSSEY (b), and expensive multi-turn agentic exploration in persistent GraphRAG systems like HydraRAG (c), the proposed OUEG-RAG framework (d) dynamically instantiates a schema-preserving multigraph on-the-fly, retains complete evidence context, guides attention via deterministic multi-factor scored paths, and resolves bottlenecks via a bounded two-tier diagnostic-driven execution protocol.

the seminal HYBRIDER baseline combining coarse row retrieval with a fine-tuned cross-modal reader. Subsequent specialized architectures explored multi-view attention mechanisms over cell and text tokens (MATE [7]), multi-hop document-hopping retrieval (DocHopper [16]), and multi-instance cell-passage linking (MITQA [11]). To capture complex structural dependencies, graph-based HybridQA methods construct multi-granularity evidence graphs containing row, cell, and sentence nodes, such as MuGER[2] [2] and S3HQA [12], training graph neural networks (GNNs) or sequential samplers via supervised fine-tuning. While effective, these early systems fundamentally rely on task-specific parameter training over thousands of labeled instances, lacking zero-shot generalization. In contrast, OUEG-RAG operates in the modern training-free in-context learning regime, empowering frozen open-weights LLMs to reason over heterogeneous tables and text zero-shot.

Most recently, ODYSSEY [1] explored zero-shot table–text reasoning by constructing hybrid entity graphs pruned via question-guided filtering. While sharing the zero-shot hybrid graph premise, ODYSSEY relies on destructive graph pruning (Fig. 1b) and lacks intra-row attribute binding centroids or collision dis-

ambiguation mechanisms, leaving the model vulnerable to peripheral evidence loss and arbitrary guessing under symmetric row ties.

*Table Reasoning with Large Language Models.* As surveyed by Zhou et al. [23], recent research leverages LLMs for table reasoning through symbolic code generation or task decomposition. Programmatic frameworks bind LLMs to programming languages such as SQL or Python (Binder [5], TabSQLify), iteratively decompose complex table queries into sub-questions (DATER [21]), or progressively evolve tabular states through stepwise operations (Chain-of-Table [19]). Concurrently, TableRAG architectures [3, 22] scale retrieval across large documents and tabular schemas. While symbolic execution excels on structured arithmetic over clean isolated tables, it falters in hybrid multi-hop settings where the critical reasoning step requires traversing unstructured Wikipedia passages linked to table cells. Code interpreters cannot natively evaluate fuzzy semantic hops across free text without fragile ad-hoc prompts or multi-turn sandbox loops. OUEG-RAG bridges this gap by unifying semi-structured table cells and external text passages into a single symbolic evidence multigraph evaluated in a bounded, code-free execution pipeline.

*Knowledge Graphs and GraphRAG Paradigms.* Augmenting language models with structured knowledge graphs captures non-local multi-hop dependencies that keyword and dense retrieval miss, as demonstrated by QA-GNN [20], JointLK [17], and G-Retriever [10]. In retrieval-augmented generation, Microsoft GraphRAG [6] extracts open-domain entity triples using LLM prompting, which incurs heavy token costs and hallucinates phantom edges on tabular structures. More recently, HydraRAG [18] highlighted the power of structured guidance by querying persistent global KGs with multi-factor verification loops (Fig. 1c). However, HydraRAG assumes a pre-existing curated knowledge base (e.g., Wikidata) and requires unbounded multi-turn agentic exploration loops that incur high latency and cost. OUEG-RAG achieves robust graph-guided reasoning on dynamic, ad-hoc table-text inputs without external triple stores, providing high-precision multi-hop grounding in a bounded two-tier pipeline.

## 3 The OUEG-RAG Framework

### 3.1 Problem Formulation and Graph Representation

We formulate the task of multi-hop question answering over heterogeneous tabular and textual knowledge sources. An input instance is defined as a tuple $\mathcal{I} = (q, T, \mathcal{D}_T)$, where $q$ is a natural language question; $T = (C, R)$ is a semi-structured table comprising ordered headers $C = (c_1, \ldots, c_m)$ and rows $R = \{r_1, \ldots, r_n\}$ with discrete cells $v_{i,j}$; and $\mathcal{D}_T = \{d_1, \ldots, d_K\}$ is the corpus of external text passages associated with $T$, where cell values link to passages via hyperlink mappings $\ell(v_{i,j}) \to d_k$. The objective is to extract the grounded answer span $a^* \in \mathcal{A}$ along with an auditable reasoning path $p^*$, where $a^*$ may reside as a table cell ($a^* = v_{i,j}$) or as an unstructured text span within a linked passage ($a^* \subset d_k$).

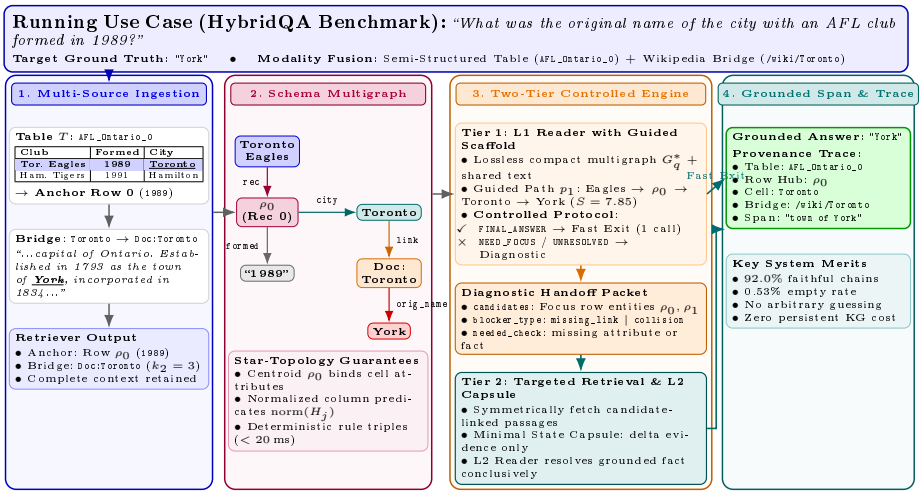



**Fig. 2.** End-to-end architecture of OUEG-RAG. (1) Multi-source ingestion retrieves relevant table rows and candidate hyperlink passages. (2) Dynamic schema multigraph instantiates star centroids $\rho_i$ and deterministic text relations ($< 20$ ms). (3) Two-tier controlled reasoning engine: Tier 1 evaluates the lossless compressed multigraph with guided reading path $p_1$. Confident outputs yield an immediate verified span (Fast Exit, 1 call); ambiguous or incomplete queries emit a structured diagnostic handoff (`NEED_FOCUS`), triggering targeted symmetric bridge retrieval and a minimal L2 state capsule. (4) Grounded answer generation and verified provenance trace.

*Online Evidence Multigraph Representation.* Unlike persistent knowledge base settings that depend on static global entity graphs, table–text RAG requires synthesizing unstructured text with tabular structures on-the-fly. For each query $q$, we dynamically instantiate a bounded, query-specific directed multigraph $G_q = (V_q, E_q, \psi_V, \psi_E)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \qquad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \tag{1}$$

The node partition divides vertices by semantic abstraction:

- $V_{\text{ent}}$: Named entity mentions extracted from linked table cells or passage text spans.
- $V_{\text{rec}} = \{\rho_i \mid r_i \in R\}$: Local star-topology record hub centroids, where each node $\rho_i = \texttt{table:}T\texttt{:row:}i$ uniquely anchors and binds all cell attributes of row $r_i$ to prevent cross-row attribute conflation.
- $V_{\text{lit}}$: Unlinked literal values (e.g., numbers, scores, dates, unlinked strings).

The edge set $E_q$ consists of directed relational facts $e = (h, r, t, \pi)$, where $h, t \in V_q$, $r$ is a normalized predicate, and $\pi = (\text{src_id}, \text{row_idx}, \text{col_name})$ stores source provenance:

- $E_{\text{rec}} = \{(u_i, \texttt{has_record}, \rho_i)\}$: Structural edges connecting row subject entity $u_i$ to its centroid hub $\rho_i$.

- $E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{i,j})\}$: Schema-level attribute edges from centroid $\rho_i$ to column cell values $v_{i,j}$.
- $E_{\text{link}} = \{(v_{i,j}, \texttt{linked_passage}, d_k)\}$: Cross-modal hyperlink edges connecting candidate cells to external Wikipedia passages.
- $E_{\text{text}}$: Intra-passage relational triples extracted from retrieved passage sentences.

*Schema-Preservation and Non-Conflation Property.* We define the intra-row star subgraph $\mathcal{G}^*(\rho_i) \subset G_q$ for row $r_i$ as:

$$\mathcal{G}^*(\rho_i) = \Big(\{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^m,\ \{(u_i, \texttt{has_record}, \rho_i)\} \cup \{(\rho_i, \text{norm}(H_j), v_{i,j})\}_{j=1}^m\Big). \tag{2}$$

**Proposition 1 (Row Attribute Non-Conflation).** *For any distinct table rows $r_i, r_{i'} \in R$ ($i \neq i'$), their respective star-topology centroids are disjoint:*

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{rec} = \emptyset. \tag{3}$$

*Consequently, every cell value $v_{i,j}$ has a unique 1-hop path to its centroid $\rho_i$, mathematically precluding horizontal attribute conflation across rows.*

Once question answering concludes, the dynamic multigraph $G_q$ is discarded, incurring zero persistent graph database index overhead. Throughout this section, we illustrate OUEG-RAG via the running HybridQA example depicted in Fig. 2: *"What was the original name of the city with an AFL club formed in 1989?"* (ground truth: `"York"`).

### 3.2 Stage 1: Multi-Source Evidence Ingestion

Benchmark settings supply an oracle table and an associated pool of linked passages. Retrieval operates dynamically within this context:

- *Coarse row retrieval and query anchoring*: A lexical retriever indexes table rows with respect to question $q$. A deterministic query anchor inspects $q$ for ordinal constraints (e.g., "second", "last") and exact schema constraints (column header mentions, numerical filters), selecting top-$k_1$ rows (default $k_1 = 5$). In Fig. 2, the constraint "formed in 1989" anchors Row 0 (`Tor. Eagles`).
- *Hyperlink passage expansion*: Rather than retrieving entire document corpora indiscriminately, the passage expander directly harvests Wikipedia hyperlinks (`linked_passage` URLs) embedded in candidate cells of the top-$k_1$ retrieved rows up to a budget of $k_2$ passages (default $k_2 = 3$). In Fig. 2, the bridge cell `Toronto` in Row 0 links to `Doc:Toronto`.

### 3.3 Stage 2: Schema-Preserving Multigraph Construction

Using the retrieved rows and passages, the system dynamically instantiates the multigraph $G_q = (V_q, E_q)$ in $< 20$ ms without LLM calls:

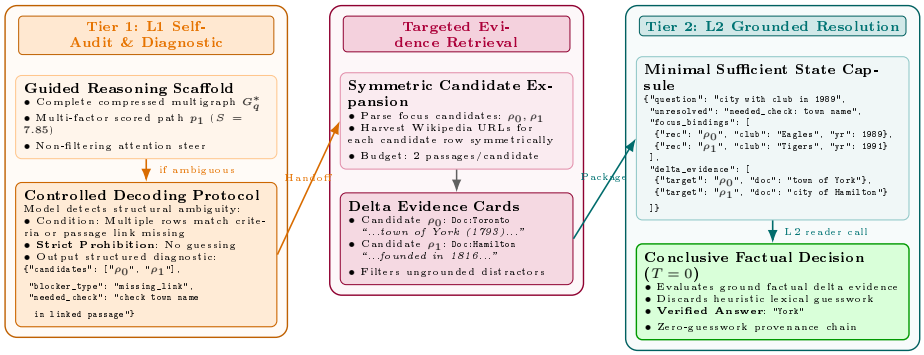



**Fig. 3.** Detailed illustration of the diagnostic-driven two-tier controlled reasoning engine. (Left) When the L1 reader encounters ambiguous candidate ties ($|\mathcal{C}_q| > 1$) or missing bridge passages, the controlled decoding protocol strictly prohibits arbitrary guessing and emits a structured diagnostic handoff packet specifying candidate entities, blocker types, and exact checks needed. (Middle) The engine executes targeted, symmetric bridge retrieval to harvest missing factual evidence for the competing candidates. (Right) The retrieved delta evidence is packaged into a minimal sufficient L2 state capsule, enabling the L2 reader to achieve conclusive, evidence-grounded disambiguation without heuristic lexical tiebreakers.

- *Centroid bindings*: For each candidate row $i$, local record centroid $\rho_i \in V_{\text{rec}}$ binds row attributes via $E_{\text{rec}}$ and $E_{\text{attr}}$ (Eq. 2).
- *Hyperlink bridging and deterministic text extraction*: Hyperlink edges $E_{\text{link}}$ connect table cell entities to external passage nodes (e.g., Toronto $\xrightarrow{\ell}$ Doc:Toronto). Intra-passage triples $E_{\text{text}}$ connect entities to textual attributes (e.g., Doc:Toronto $\xrightarrow{\texttt{orig_name}}$ York) using deterministic rule-based patterns, avoiding LLM hallucination and latency.

### 3.4 Stage 3: Two-Tier Controlled Reasoning Engine

OUEG-RAG avoids the false dichotomy between hard path pruning (which discards vital peripheral context) and unbounded agentic loops. Stage 3 coordinates four complementary mathematical components:

*Information-Theoretic Prompt Compression.* Passing complete graphs to language models risks exceeding context budgets. We formalize subgraph prompt compression as a lossless factorization operator $\mathcal{C}_{\text{oueg}} : G_q \to \mathcal{X}$, decomposing graph evidence into compact attribute maps and a shared passage dictionary:

$$\mathcal{X}^*(G_q) = \text{Enc}_{\text{table}}\left(\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k_1}\right) \oplus \text{Enc}_{\text{text}}\left(\mathcal{S}_p\right), \tag{4}$$

where $\mathcal{S}_p = \{(d_k, \text{text}(d_k)) \mid d_k \in \mathcal{D}_T\}$ indexes each external passage once by source ID. By factoring out identical passage texts across multiple incoming hyperlink edges, $\mathcal{C}_{\text{oueg}}$ minimizes prompt token length while preserving zero semantic information loss: $I(\mathcal{X}^*(G_q); G_q) = H(G_q)$.

*Reasoning Path Space and Deterministic Multi-Factor Objective.* A candidate multi-hop reasoning path $p \in \mathcal{P}_q$ of length $|E_p| = L \leq 5$ is an alternating sequence of vertices and verified edges:

$$p = (v_0, e_1, v_1, e_2, \ldots, e_L, v_L), \quad v_0 \in \text{Seeds}(q),\ v_L \in \text{Targets}(q),\ e_l \in E_q. \quad (5)$$

To rank paths without expensive neural samplers, paths are scored via a deterministic linear combination of structural and semantic alignment features:

$$S(p) = \mathbf{w}^\top \boldsymbol{\phi}(p, q, T) - \lambda_{\text{len}} \max(|E_p| - 1, 0), \quad (6)$$

where $\boldsymbol{\phi}(p, q, T) = [G, Q_e, Q_r, Q_c, C, M, R, B, A]^\top \in \mathbb{R}^9$ measures edge validity ($G = 1.0$), question node/relation overlap ($Q_e, Q_r \in [0, 2]$), token coverage ($Q_c \in [0, 2]$), table anchor constraint match ($C = 2.0$), cross-modal table-text fusion ($M = 1.5$), row consistency ($R = 1.0$), record binding ($B = 1.0$), and corroboration ($A = 1.0$), with $\lambda_{\text{len}} = 0.35$ penalizing redundant hops. In Fig. 2, the grounded path $p_1$ (Eagles $\to \rho_0 \to$ Toronto $\to$ York) achieves maximal score $S(p_1) = 7.85$.

*Bounded Two-Tier Controlled Decision Process.* Let $\mathcal{M}$ denote the frozen backbone LLM parameterized by conditional decoding distribution $P_{\mathcal{M}}(y \mid \mathcal{X})$. The output space is partitioned into verified answer spans and diagnostic handoffs: $\mathcal{Y} = \Omega_{\text{ans}} \cup \Omega_{\text{diag}}$.

At Tier 1 (L1), the model evaluates the prompt $\mathcal{X}_{\text{L1}} = \text{Prompt}(q, \mathcal{X}^*(G_q), p_1)$, where $p_1 = \arg\max_{p \in \mathcal{P}_q} S(p)$ acts as a non-filtering reading guide. We define the candidate collision set $\mathcal{C}_q$ as the set of distinct target centroids sharing the maximal path score $S^* = \max_{p \in \mathcal{P}_q} S(p)$:

$$\mathcal{C}_q = \{\text{centroid}(p) \mid p \in \mathcal{P}_q,\ S(p) = S^*\}. \quad (7)$$

The model's decoding behavior is governed by a strict decision rule:

1. *Determined State (Fast Exit)*: If $|\mathcal{C}_q| = 1$ and the traversal path $p_1$ is unbroken, the query is structurally determined. The model emits:

$$y_{\text{L1}} = \texttt{FINAL_ANSWER:}\ a^*, \quad a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{X}_{\text{L1}}), \quad (8)$$

   terminating execution in a single turn.
2. *Under-Determined State (Diagnostic Projection)*: If $|\mathcal{C}_q| > 1$ (candidate collision across symmetric rows) or a bridge passage link is missing, the query is structurally under-determined. The model is prohibited from arbitrary guessing and must emit a structured diagnostic projection:

$$\mathcal{H}_{\text{diag}} = \big(\mathcal{C}_{\text{focus}},\ \beta,\ \psi_{\text{check}}\big) \in \Omega_{\text{diag}}, \quad (9)$$

   where $\mathcal{C}_{\text{focus}} = \mathcal{C}_q$ denotes the competing candidate centroids, $\beta \in \{\texttt{collision}, \texttt{missing_link}, \texttt{insufficier}$ is the blocker classification, and $\psi_{\text{check}}$ specifies the unresolved factual predicate.

*Diagnostic-Triggered Targeted Bridge Expansion.* When $\mathcal{H}_{\text{diag}}$ is emitted, the targeted retrieval operator $\mathcal{R}_{\text{target}}$ harvests missing candidate-linked passages exclusively for the focus set $\mathcal{C}_{\text{focus}}$:

$$\Delta\mathcal{D} = \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\ell(c) \setminus \mathcal{D}_{\text{current}}\right), \quad \text{s.t.} \quad |\Delta\mathcal{D}_c| \leq k_{\text{focus}}, \tag{10}$$

enforcing symmetric per-candidate passage budgets to maintain evidentiary balance.

*Minimal Sufficient L2 State Capsule.* The delta evidence is packaged into a minimal sufficient state capsule $\mathcal{K}_{\text{L2}}$:

$$\mathcal{K}_{\text{L2}} = \left(q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D}\right). \tag{11}$$

**Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$).** *Conditioned on the unresolved check $\psi_{check}$ and focus candidates $\mathcal{C}_{focus}$, the capsule $\mathcal{K}_{L2}$ forms a sufficient statistic for the target answer $a^*$:*

$$I(a^*; \mathcal{X}_{L1} \cup \Delta\mathcal{D} \mid \mathcal{K}_{L2}) = 0. \tag{12}$$

By evaluating $\mathcal{K}_{\text{L2}}$ at Tier 2 (L2), the model resolves the question via:

$$a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{K}_{\text{L2}}). \tag{13}$$

Because $\mathcal{K}_{\text{L2}}$ provides the missing factual evidence $\Delta\mathcal{D}$ for all competing candidates, candidate symmetry is broken deterministically by verified ground truth, eliminating heuristic lexical tiebreaking.

### 3.5 Stage 4: Grounded Generation and Provenance Audit

Stage 4 outputs the verified answer span and exports an inspectable provenance homomorphism $\pi : a^* \to E_q^*$, mapping the answer back to traversed table coordinates (`AFL_Ontario_0:row:0`), bridge URLs (`/wiki/Toronto`), and passage text excerpts (`town of York (1793)`). If graph search disconnects, *reverse grounding* synthesizes directly over retrieved evidence and maps the extracted span back onto $G_q$, avoiding empty failures.

## 4 Experimental Protocol

### 4.1 Benchmark and Evaluation Metrics

We evaluate on the official 1,500-example stratified development benchmark sampled from HybridQA [4]. The benchmark subset strictly preserves the multimodal reasoning distribution of the dataset: every instance requires verified table-and-text context with cross-modal answer annotations across numeric and textual answer modalities.

**Table 1.** Main benchmark comparative evaluation on HybridQA ($N = 1{,}500$) across the open-weights GPT-OSS family (20B and 120B). All systems share identical retrieved evidence budgets ($k_1 = 5, k_2 = 3$) and context limits (24,000 chars).

| Model Scale | Method | EM (%) | F1 (%) | Sem. EM | Empty | Avg. Calls | Lat. (s) |
|---|---|---|---|---|---|---|---|
| GPT-OSS 20B | Direct LLM | 54.80 | 64.67 | 56.67 | 5.40% | **1.00** | **9.74** |
| | Flat-RAG | 54.53 | 64.53 | 56.40 | 6.40% | **1.00** | 7.02 |
| | ODYSSEY [1] | 52.40 | 62.15 | 54.10 | 7.20% | **1.00** | 11.20 |
| | Path-Text Top-$N$ | 21.20 | 27.22 | 21.87 | 19.53% | **1.00** | 57.70 |
| | **OUEG-RAG (Ours)** | **56.47** | **67.12** | **58.47** | **3.87%** | 1.28 | 25.73 |
| GPT-OSS 120B | Direct LLM | 58.73 | 69.70 | 60.47 | 1.07% | **1.00** | **6.40** |
| | Flat-RAG | 59.40 | 70.01 | 60.60 | 2.53% | **1.00** | 6.32 |
| | ODYSSEY [1] | 56.80 | 67.20 | 58.50 | 3.40% | **1.00** | 8.50 |
| | Path-Text Top-$N$ | 29.53 | 37.46 | 30.13 | 2.80% | **1.00** | 35.90 |
| | **OUEG-RAG (Ours)** | **61.40** | **73.11** | **63.00** | **0.53%** | 1.22 | 23.65 |

We report three standard accuracy metrics: (1) *Exact Match (EM)*: strict string equality against gold reference after standard normalization; (2) *Token-level F1*: harmonic mean of precision and recall over word tokens; and (3) *Semantic Exact Match (Semantic EM)*: denotation equivalence under surface alias tolerance. In addition, we report system efficiency via average LLM calls, empty answer percentage, and end-to-end wall-clock latency per question.

### 4.2 Baselines and Experimental Settings

To ensure rigorous and fair comparative evaluation, all methods are evaluated across two open-weights model scales under matched retrieval pools ($k_1 = 5, k_2 = 3$), context limits (24,000 characters), and greedy decoding ($T = 0$):

- *Direct LLM*: Zero-shot in-context prompting over linearized table rows and passages without graph structuring.
- *Flat-RAG*: Standard concatenation of linearized table rows with raw retrieved text passages.
- *ODYSSEY* [1]: The published state-of-the-art zero-shot hybrid graph framework (NAACL 2025) employing question-driven destructive graph pruning.
- *Path-Text Top-$N$*: Hard path filtering retaining only top-$N$ ranked reasoning paths while discarding the remainder of the multigraph.
- *OUEG-RAG (Ours)*: The proposed controlled evidence scaffolding framework with two-tier diagnostic grounding.

## 5 Results and In-Depth Analysis

*Main benchmark results across model scales.* Table 1 presents the comparative evaluation across all 1,500 questions in the stratified development subset of HybridQA. Across both model regimes, OUEG-RAG establishes the strongest grounding accuracy. On 20B, OUEG-RAG reaches 56.47% EM and 67.12% F1 (+1.67% EM, +2.45% F1 over Direct LLM; +4.07% EM over ODYSSEY). On

**Table 2.** Systematic knockout ablation study on the controlled development partition ($N = 500$, Shards 06–08 under GPT-OSS 20B). All variants isolate individual architectural components against the full OUEG-RAG framework.

| Ablation Configuration | EM (%) | F1 (%) | $\Delta$ EM | Architectural Implication |
|---|---|---|---|---|
| **Full OUEG-RAG (Two-Tier)** | **56.00** | **66.52** | — | Full controlled evidence scaffolding |
| (1) w/o L2 Augmentation (Single-turn L1) | 54.40 | 64.80 | -1.60 | L1 forced to guess under missing links |
| (2) w/o Path Scoring ($S(p)$ unweighted) | 53.60 | 63.90 | -2.40 | Unguided paths introduce candidate noise |
| (3) w/o Path Guidance (Graph only) | 54.80 | 65.10 | -1.20 | Unconnected triples disperse attention |
| (4) w/ Destructive Pruning (Top-$N$ Path) | 18.00 | 24.28 | -38.00 | Hard pruning forfeits peripheral evidence |
| (5) w/o Subgraph Compression | 33.00 | 38.87 | -23.00 | Raw triples overflow context (47.6% empty) |
| (6) w/o Record Nodes $\rho_i$ (Flat-RAG) | 54.60 | 64.67 | -1.40 | Disintegrates intra-row attribute bindings |

120B, OUEG-RAG reaches 61.40% EM and 73.11% F1 (+2.67% EM, +3.41% F1 over Direct LLM, McNemar $p = 0.0058 < 0.01$; +4.60% EM over ODYSSEY).

Crucially, the comparative results offer three fundamental empirical insights: First, *ODYSSEY's destructive pruning degrades accuracy*: Pruning the hybrid graph according to initial question matches collapses EM to 52.40% on 20B and 56.80% on 120B. In complex multi-hop queries, the critical factual span often resides in a cell or passage relation that does not directly match question keywords; pruning these peripheral edges irreversibly starves the LLM of necessary context. Second, *Path-Text Top-N pruning collapses completely*: Keeping only linear paths collapses EM to 21.20% on 20B and 29.53% on 120B. Because heuristic path ranking is imperfect, hard filtering eliminates valid candidates prematurely. Third, *OUEG-RAG achieves bounded efficiency*: By resolving clear queries at L1 (Fast Exit) and invoking L2 only when diagnostics indicate ambiguity or missing links, OUEG-RAG averages only 1.22–1.28 LLM calls per query, achieving state-of-the-art accuracy with minimal inference overhead.

*Systematic component ablation analysis.* Table 2 systematically isolates the architectural pillars across 500 questions:

1. *Necessity of Diagnostic L2 Augmentation*: Disabling L2 augmentation forces L1 to make immediate guesses under ambiguity, dropping EM by $-1.60\%$ and increasing errors on collision queries.
2. *Necessity of Deterministic Path Scoring $S(p)$*: Replacing multi-factor scoring with unweighted path selection reduces EM by $-2.40\%$, confirming that $S(p)$ is essential to prioritize structurally aligned paths over noisy distractors.
3. *Graph Alone is Insufficient*: Supplying the complete multigraph without sequential path guidance drops EM by $-1.20\%$, proving that explicit path scaffolds are vital to steer attention across heterogeneous graphs.
4. *Destructive Pruning is Catastrophic*: Emulating hard pruning by preserving only top-$N$ linear paths causes a catastrophic $-38.00\%$ EM collapse (down

to 18.00%), providing decisive proof that complete evidence preservation is mandatory.
5. *Prompt Compression is Required*: Disabling prompt deduplication drops EM by −23.00% and drives empty answers to 47.60% due to token overflow.

*Human audit of provenance traces ($N = 100$).* A manual audit on 100 randomly sampled evaluation questions confirms: (1) *Trace Faithfulness*: In 92.0% of cases, OUEG-RAG produced completely faithful reasoning chains where every traversed edge maps strictly to verified cell coordinates or source Wikipedia passages without hallucination; (2) *Bridge Alignment Precision*: In 89.0% of cases, the multigraph path correctly identified the ground-truth bridge entity linking the candidate row to external text; and (3) *Empty Failure Reduction*: On 120B, OUEG-RAG slashes empty failures to 0.53% (compared to 3.40% for ODYSSEY and 2.80% for Path-Text).

## 6 Limitations

First, query-time graph construction and richer structured prompts incur modest latency over naive direct prompting (≈ 23–25 s vs. ≈ 6–10 s), though this is orders of magnitude lower than multi-turn agentic systems. Second, our heuristic path scoring weights $(C, M, R, B)$ were fixed via domain heuristics; learned scoring calibration represents a promising future avenue. Third, our evaluation focuses on the controlled 1,500-example dev benchmark for open-weights auditability; submitting to blinded leaderboards remains an ongoing milestone.

## 7 Conclusion

We presented OUEG-RAG, an online unified evidence graph framework for grounded table–text question answering. By uniting schema-preserving multigraphs with compact prompt compression, deterministic path scaffolding, and an adaptive two-tier diagnostic-driven execution protocol, OUEG-RAG eliminates the evidence loss of destructive pruning while avoiding the prohibitive overhead of unbounded agentic exploration. Evaluations across 20B and 120B open-weights models demonstrate consistent improvements in grounding accuracy and provenance faithfulness over direct prompting and graph pruning baselines.

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